

EEE 359

Engineering Project Management

Lecture 03

Investment Appraisal

Investment Appraisal

The investment appraisal improve the decision-making process regarding the desirability or viability of the project. It should examine all the realistic options before making a firm recommendation for the proposed case. The investment appraisal must also include a cost/benefit analysis and take into account all the relevant factors such as:

- Capital costs, operating costs and overhead costs
- Support and training costs
- Dismantling and disposal costs
- Expected residual value (if any)
- Any cost savings that the project will bring
- Any benefits that cannot be expressed in monetary terms

Return on Investment (ROI)

- Return on Investment (ROI) measures the gain or loss of an investment relative to the amount of money invested.
- ROI is usually expressed as a percentage.
- ROI formula is: $ROI = (\text{Net Profit} / \text{Cost of Investment}) \times 100$
- When comparing the ROI of two projects - **Bigger is Better**

Return on Investment (ROI)

For monetary returns,

$$ROI = [(Income - Investment) / Investment] * 100$$

For marketing investments,

$$ROI = \left(\frac{Sales\ increase -\ campaing\ cost}{campaing\ cost} \right) \cdot 100$$

For technological projects,

$$ROI = \left(\frac{incremental\ benefits -\ software\ cost}{software\ cost} \right) \cdot 100$$

For human resource development projects,

$$ROI = \left(\frac{Productivity\ increase -\ Training\ Cost}{Training\ Cost} \right) \cdot 100$$

Return on Investment (ROI)

A Company invests \$10,000 in a project and receives \$12,000 two years later.

What is the ROI?

Net profit = (\$12000 - \$10000) = \$2000

ROI = (\$2000/\$10000) $\times$ 100 = 20%

Present Value (PV)

Present Value (PV) is the current value of a future amount of money, discounted back using a required rate of return (discount rate).

Money today is worth more than money in the future because:

- It can earn interest
- Inflation reduces purchasing power
- There is risk

$$PV = \frac{FV}{(1 + r)^t}$$

FV = Future Value, **r** = Interest Rate/Discount Rate, **t** = Number of Periods

Present Value (PV)

Present value (PV) is the value today of future cash flows.

If you wanted to have \$10,000 in 3 years, what amount of money would you need today to get this amount if the money was earning 7%?

$$\mathbf{PV = FV/(1+r)^n}$$

$$PV = \$10000/(1+.07)^3$$

$$PV = \$10000/(1.07*1.07*1.07)$$

$$PV = \$10000/1.225$$

$$PV = \$8163.26$$

FV = Future Value, r = Interest Rate, n = Number of Periods

Net Present Value (NPV)

Net Present Value (NPV) is the present value of all future cash flows (both inflows and outflows) minus the initial investment.

“After accounting for time value of money, how much value does this project create?”

$$NPV = \sum \frac{CF_t}{(1 + r)^t} - \text{Initial Investment}$$

CF_t = Cash flow at time t , r = Interest Rate/Discount rate, t = Time period

Decision:

- $NPV > 0 \rightarrow$ Accept project
- $NPV = 0 \rightarrow$ Indifferent
- $NPV < 0 \rightarrow$ Reject project

Net Present Value (NPV)

Net Present Value (NPV) compares the value of a dollar today vs the value of that same dollar in the future after taking inflation and return into account.

When comparing the NPV of two projects - **Bigger is Better**

Project A has a NPV of \$150,000 and takes six months or Project B having of \$295,000 and takes one year.

Which project do you choose?

Net Present Value (NPV)

Year	Income in \$	Discount Rate (%)
1	10000	5
2	11000	5
3	12000	5
4	12000	5
Total	45000	

What is the NPV of this project? Assume total initial investment is \$30,000

Net Present Value (NPV)

Year	Income in \$	Discount Rate (%)	Discount Factor
1	10000	5	$1/1.05 = 0.9523$
2	11000	5	$1/(1.05)^2 = 0.9070$
3	12000	5	$1/(1.05)^3 = 0.8638$
4	12000	5	$1/(1.05)^4 = 0.8227$
Total	45000		

Net Present Value (NPV)

Year	Income in \$	Discount Rate (%)	Discount Factor	NPV in \$
1	10000	5	$1/1.05 = 0.9523$	$10000 \times 0.9523 = 9523.8$
2	11000	5	$1/(1.05)^2 = 0.9070$	$11000 \times 0.9070 = 9977$
3	12000	5	$1/(1.05)^3 = 0.8638$	$12000 \times 0.8638 = 10365.6$
4	12000	5	$1/(1.05)^4 = 0.8227$	$12000 \times 0.8227 = 9872.4$
Total	45000			39788.8

Payback Period

Payback is the time taken to recover the capital outlay of the project, having taken into account all the operating and overhead costs during this period. Usually, this is based on the undiscounted cash flow.

Payback is easily calculated by summing all the net incomes until the total equals the original investment.

If the original investment is \$600,000, and the net income is \$75,000 per year for the next 10 years, the payback is $600,000 / 75,000 = 8$ years

When comparing the Payback period of two projects - **Smaller is Best**

Internal Rate of Return (IRR)

Interest rate at which the project inflows (revenues) and project outflows (costs) are equal. In other words, it is the discount rate at which the NPV is 0.

The bigger the IRR, the better.

Internal Rate of Return (IRR)

$$NPV = \sum \frac{CF_t}{(1+r)^t} - \text{Initial Investment} = 0$$

A project investment has an initial investment of \$100,000. Expected cash flows:

Year	Cash Flow
1	\$40,000
2	\$40,000
3	\$40,000

What is the IRR?

$$0 = \frac{40,000}{(1+r)^1} + \frac{40,000}{(1+r)^2} + \frac{40,000}{(1+r)^3} - 100,000 \quad \longrightarrow \quad \boxed{\text{Solve for } r}$$

Summary of Investment Appraisal

- NPV = summation of PVs – original investment
- Net income = incoming money- – outgoing money.
- Payback period = no. of years it takes for net income to equal original investment
- Profit = total net income – original investment
- Average return/annum = total net income/no. of years
- Return on investment (%)
$$ROI = [(Income - Investment) / Investment] * 100$$
- IRR = % discount rate for NPV = 0

Cost/Benefits Analysis

- Benefit Cost Ratio (BCR) compares the benefit to the cost of the initiative. The format is 4:1, which means the benefits of the project outweigh the costs 4 to 1.
- Remember **Profit is not the same as Benefits/Revenue**. This is simply noise. Benefit and cost are the main factors.
- A project can have a BCR of less than one (.75:1) or a benefit of \$.75 for each \$1.00 invested. You would not approve the project unless there was an underlying factor.

What is the benefit cost ratio if revenue=\$200,000 and cost=\$50,000?

$$\$200,000 / \$50,000 = 4$$

Then the BCR is 4:1

Cost/Benefits Analysis

Following is a list of some of the benefits that have to be considered, from which it will be apparent that some will be very difficult to quantify in monetary terms:

- Financial
- Statutory
- Economy
- Risk reduction
- Productivity
- Reliability
- Staff morale
- Cost reduction
- Safety
- Flexibility
- Quality
- Delivery
- Social

Thanks